Abstract Algebra | Quiz 9 | November 4, 2026

Name: Richard | Hammack | Score: _____

1. (3 points) Explain why $\mathbb{Z}_{24} \cong \mathbb{Z}_3 \times \mathbb{Z}_8$.

$\mathbb{Z}_{24} = \langle 1 \rangle$, and $\mathbb{Z}_3 \times \mathbb{Z}_8 = \langle (1,1) \rangle$, so both groups are cyclic of the same order, so they are isomorphic.

2. (3 points) Explain why $\mathbb{Z}_{24} \ncong \mathbb{Z}_4 \times \mathbb{Z}_6$.

Any element of $\mathbb{Z}_4 \times \mathbb{Z}_6$ has an order that divides $\text{lcm}(4,6) = 12$. Thus no element of $\mathbb{Z}_4 \times \mathbb{Z}_6$ generates the entire group, so $\mathbb{Z}_4 \times \mathbb{Z}_6$ is not cyclic, but $\mathbb{Z}_{24}$ is cyclic.

3. (6 points) List all abelian groups of order 360, up to isomorphism.

$360 = 8 \cdot 5 \cdot 9 = 2^3 \cdot 5^1 \cdot 3^2$

- $\mathbb{Z}_8 \times \mathbb{Z}_5 \times \mathbb{Z}_9 \quad (\cong \mathbb{Z}_{360})$
- $\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_9$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9$
- $\mathbb{Z}_8 \times \mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_3$
- $\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_3$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_3 \times \mathbb{Z}_3$